

Real-Time SAR Ship Detection: A Speckle-Model-Based DSP Pipeline

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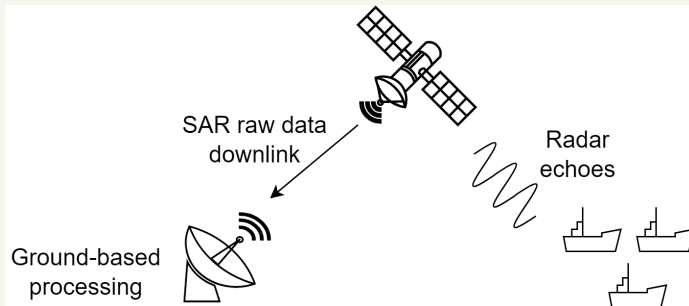
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- ▶ Previous Methods
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Introduction

- ▶ SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- ▶ Detected ships provide information for maritime surveillance.



Problem and Motivation

- ▶ Ground-Based Processing
 - ▶ Large raw-data volume
 - ▶ Limited downlink bandwidth
 - ▶ Delayed target information
- ▶ Onboard Processing
 - ▶ Reduced data transmission and detection latency
 - ▶ Challenge: Limited computation, memory, and power
- ▶ Research Goal
 - ▶ Lightweight SAR processing for low-latency ship detection